Lab 9: 12% of cases skip get_customer: fix it with a gate, not a sentence

Topic 1: Agentic Architecture & Orchestration · Task 1.4: Implement multi-step workflows with enforcement and handoff patterns

Runs on Simulator 1: Customer Support Resolution Agent · Run in class

Source: Official exam guide, sample question 1

Commit to an answer first

Production data shows that in 12% of cases, your agent skips get_customer entirely and calls lookup_order using only the customer's stated name, occasionally leading to misidentified accounts and incorrect refunds. What change would most effectively address this reliability issue?

A. Add a programmatic prerequisite that blocks lookup_order and process_refund calls until get_customer has returned a verified customer ID.

B. Enhance the system prompt to state that customer verification via get_customer is mandatory before any order operations.

C. Add few-shot examples showing the agent always calling get_customer first, even when customers volunteer order details.

D. Implement a routing classifier that analyzes each request and enables only the subset of tools appropriate for that request type.

Simulate the solution

```
# In the browser: open Simulator 1, then Lab 9
cd labs
uv run serve.py

# In Claude Code, from the labs folder
/lab 9

# In a terminal, from simulators/simulator-1-customer-support/
uv run support_agent.py --case CS-03
uv run support_agent.py --case CS-03 --no-gate --tag nogate
```

What you should see

With the gate on, get_customer always comes first. Compare the two traces.

Open `simulators/simulator-1-customer-support/evidence/` and find it yourself.

The answer

A. A prerequisite hook blocks lookup_order and process_refund until get_customer returns an id.

When a specific tool sequence is required for critical business logic (like verifying customer identity before processing refunds), programmatic enforcement provides deterministic guarantees that prompt-based approaches cannot. Options B and C rely on probabilistic LLM compliance, which is insufficient when errors have financial consequences. Option D addresses tool availability rather than tool ordering, which is not the actual problem.

Now change one thing

Copy a file in cases/ to a new CS id, change the customer's message and expected disposition, rerun with --case, then score it with evaluate.py.